

$$i_{112} \rightarrow$$

$$\begin{aligned} & \frac{1}{16} \left(-\frac{1}{6} g^4 Y_d^{i112} L_{F3,0}[m_2] - \frac{1}{6} g^4 Y_d^{i112} L_{F4,-1}[m_2] + \frac{8}{45} g^4 Y_d^{i112} L_{F5,-2}[m_2] + \frac{8}{72} g^2 Y_d^{i112} \right. \\ & \quad \left. (18 Y_{ePR} Y_{ePR} - 13 \sum g^2) L_{F3,0}[m_P] + \frac{1}{48} g^2 Y_d^{i112} (-12 Y_{ePR} Y_{ePR} + 11 \sum g^2) L_{F4,-1}[m_P] - \right. \\ & \quad \left. \frac{2}{45} \sum g^4 Y_d^{i112} L_{F5,-2}[m_P] + \frac{1}{24} g^2 Y_d^{i112} (18 Y_{dPR} Y_{dPR} + 18 Y_{uPR} Y_{uPR} - 13 \sum g^2) L_{F3,0}[m_q^P] - \right. \\ & \quad \left. \frac{1}{16} g^2 Y_d^{i112} (-12 (Y_{dPR} Y_{dPR} + Y_{uPR} Y_{uPR}) + 11 \sum g^2) L_{F4,-1}[m_q^P] - \frac{2}{15} \sum g^4 Y_d^{i112} L_{F5,-2}[m_q^P] \right. \\ & \quad \left. - \frac{1}{18} g^4 Y_d^{i112} L_{F3,0}[\tilde{\mu}] - \frac{1}{12} g^2 Y_d^{i112} L_{F4,-1}[\tilde{\mu}] - \frac{1}{45} g^2 Y_d^{i112} L_{F5,-2}[\tilde{\mu}] \right) \\ & \quad - \frac{1}{36} g^2 Y_d^{i112} Y_{uPR} Y_u^{i1r} L_{F2,1,0}[m_1, m_P] + \frac{1}{18} g^2 Y_d^{i112} Y_{uPR} Y_u^{i1r} L_{F3,1,-1}[m_1, m_P] - \\ & \quad - \frac{1}{36} g^2 Y_d^{i112} Y_{uPR} Y_u^{i1r} L_{F4,1,-2}[m_1, m_q^P] + \frac{1}{144} g^2 Y_d^{i112} L_{F2,2,-1}[m_1, m_q^{i1}] - \\ & \quad - \frac{4}{9} g^2 Y_d^{i112} Y_{uPR} Y_u^{i1r} L_{F2,1,0}[m_1, m_u^r] + \frac{8}{9} g^2 Y_d^{i112} Y_{uPR} Y_u^{i1r} L_{F3,1,-1}[m_1, m_u^r] - \\ & \quad - \frac{4}{9} g^2 Y_d^{i112} Y_{uPR} Y_u^{i1r} L_{F4,1,-2}[m_1, m_u^r] - \frac{3}{4} g^2 Y_d^{i112} Y_{uPR} Y_u^{i1r} L_{F2,1,0}[m_2, m_q^P] + \\ & \quad - \frac{3}{2} g^2 Y_d^{i112} Y_{uPR} Y_u^{i1r} L_{F3,1,-1}[m_2, m_q^P] - \frac{3}{4} g^2 Y_d^{i112} Y_{uPR} Y_u^{i1r} L_{F4,1,-2}[m_2, m_q^P] - \\ & \quad - \frac{1}{16} g^2 Y_d^{i112} L_{F2,2,-1}[m_2, m_q^{i1}] + \frac{2}{3} g^2 Y_d^{i112} L_{F2,1,0}[m_2, \tilde{\mu}] - \\ & \quad - \frac{1}{2} g^4 Y_d^{i112} L_{F2,2,-1}[m_2, \tilde{\mu}] + \frac{1}{2} g^4 m_2^2 Y_d^{i112} L_{F2,2,0}[m_2, \tilde{\mu}] - \frac{1}{3} g^4 Y_d^{i112} L_{F3,1,-1}[m_2, \tilde{\mu}] + \\ & \quad - \frac{5}{9} g^4 Y_d^{i112} L_{F3,2,-2}[m_2, \tilde{\mu}] - \frac{3}{2} g^4 m_2^2 Y_d^{i112} L_{F3,2,-1}[m_2, \tilde{\mu}] - g^2 Y_d^{i112} L_{F4,3,-3}[m_2, \tilde{\mu}] + \\ & \quad + g^4 m_2^2 Y_d^{i112} L_{F3,3,-2}[m_2, \tilde{\mu}] - g^2 Y_d^{i112} L_{F4,2,-3}[m_2, \tilde{\mu}] + g^4 m_2^2 Y_d^{i112} L_{F4,3,-2}[m_2, \tilde{\mu}] - \\ & \quad - \frac{4}{3} g^3 Y_d^{i112} Y_{uPR} Y_u^{i1r} L_{F2,1,0}[m_3, m_q^P] + \frac{8}{3} g^3 Y_d^{i112} Y_{uPR} Y_u^{i1r} L_{F3,1,-1}[m_3, m_q^P] - \\ & \quad - \frac{4}{3} g^3 Y_d^{i112} Y_{uPR} Y_u^{i1r} L_{F4,1,-2}[m_3, m_q^P] + \frac{1}{3} g^2 g^3 Y_d^{i112} L_{F2,2,-1}[m_3, m_q^{i1}] - \\ & \quad - \frac{4}{9} g^3 Y_d^{i112} Y_{uPR} Y_u^{i1r} L_{F2,1,0}[m_3, m_u^r] + \frac{8}{9} g^3 Y_d^{i112} Y_{uPR} Y_u^{i1r} L_{F3,1,-1}[m_3, m_u^r] - \\ & \quad - \frac{4}{9} g^3 Y_d^{i112} Y_{uPR} Y_u^{i1r} L_{F4,1,-2}[m_3, m_u^r] + g^2 Y_d^{i112} \bar{a}_{ePR} a_{ePR} L_{F3,1,0}[m_P^r, m_e^r] - \\ & \quad - \frac{1}{2} g^2 Y_d^{i112} (7 \bar{a}_{ePR} a_{ePR} + \tilde{\mu}^2 Y_{ePR} Y_{ePR}) L_{F4,1,-1}[m_P^r, m_e^r] + \\ & \quad - \frac{1}{2} g^2 Y_d^{i112} (9 \bar{a}_{ePR} a_{ePR} + \tilde{\mu}^2 Y_{ePR} Y_{ePR}) L_{F5,1,-2}[m_P^r, m_e^r] + \\ & \quad - g^2 Y_d^{i112} \bar{a}_{dPR} a_{dPR} L_{F3,1,0}[m_d^r, m_d^r] - \frac{3}{2} g^2 Y_d^{i112} (9 \bar{a}_{dPR} a_{dPR} + \tilde{\mu}^2 Y_{dPR} Y_{dPR}) \\ & \quad - L_{F4,1,-1}[m_d^r, m_d^r] + \frac{1}{2} g^2 Y_d^{i112} L_{F2,1,0}[m_q^S, m_q^S] + \frac{4}{3} Y_{dPR} Y_d^{i112} Y_{uST} Y_u^{i1r} L_{F3,1,-1}[m_q^S, m_q^S] - \\ & \quad - \frac{1}{2} g^2 Y_d^{i112} (\bar{a}_{uPR} a_{uPR} + 4 \tilde{\mu}^2 Y_{uPR} Y_{uPR}) L_{F2,2,0}[m_q^P, m_u^r] + \\ & \quad - \frac{3}{2} g^2 Y_d^{i112} (\bar{a}_{uPR} a_{uPR} - \tilde{\mu}^2 Y_{uPR} Y_{uPR}) L_{F3,1,0}[m_q^P, m_u^r] - \\ & \quad - \frac{1}{4} g^2 Y_d^{i112} (\bar{a}_{uPR} a_{uPR} - 2 \tilde{\mu}^2 Y_{uPR} Y_{uPR}) L_{F3,2,-1}[m_q^P, m_u^r] + \\ & \quad - \frac{3}{4} g^2 Y_d^{i112} (-\bar{a}_{uPR} a_{uPR} + \tilde{\mu}^2 Y_{uPR} Y_{uPR}) L_{F4,1,-1}[m_q^P, m_u^r] - \\ & \quad - \frac{3}{4} Y_{dPR} Y_d^{i112} Y_{uST} Y_u^{i1r} L_{F2,1,0}[m_q^S, m_q^S] + \frac{3}{4} Y_{dPR} Y_d^{i112} Y_{uST} Y_u^{i1r} L_{F3,1,-1}[m_q^S, m_q^S] - \\ & \quad - \frac{1}{72} g^2 g^2 Y_d^{i112} L_{F2,1,0}[m_q^{i1}, m_1] + \frac{1}{8} g^2 Y_d^{i112} L_{F2,1,0}[m_q^{i1}, m_2] - \\ & \quad - \frac{2}{3} g^2 g^2 Y_d^{i112} L_{F2,1,0}[m_q^{i1}, m_3] + \frac{1}{4} g^2 Y_d^{i112} (\bar{a}_{uPR} a_{uPR} + 7 \tilde{\mu}^2 Y_{uPR} Y_{uPR}) L_{F3,1,0}[m_u^r, m_q^P] + \\ & \quad - \frac{1}{4} g^2 Y_d^{i112} (\bar{a}_{uPR} a_{uPR} + 4 \tilde{\mu}^2 Y_{uPR} Y_{uPR}) L_{F3,2,-1}[m_u^r, m_q^P] - \\ & \quad - \frac{4}{3} g^2 Y_d^{i112} (\bar{a}_{uPR} a_{uPR} + 7 \tilde{\mu}^2 Y_{uPR} Y_{uPR}) L_{F4,1,-1}[m_u^r, m_q^P] + \\ & \quad - \frac{1}{2} g^2 Y_d^{i112} (\bar{a}_{uPR} a_{uPR} + 7 \tilde{\mu}^2 Y_{uPR} Y_{uPR}) L_{F5,1,-2}[m_u^r, m_q^P] - \\ & \quad - \frac{3}{2} Y_{dPR} Y_d^{i112} Y_{uST} Y_u^{i1r} L_{F2,1,0}[m_u^r, m_d^r] + \frac{3}{2} Y_{dPR} Y_d^{i112} Y_{uST} Y_u^{i1r} L_{F3,1,-1}[m_u^r, m_d^r] - \\ & \quad - g^2 g^2 Y_d^{i112} L_{F3,1,-1}[\tilde{\mu}, m_1] - \frac{1}{2} g^4 \tilde{\mu}^2 Y_d^{i112} L_{F3,2,-1}[\tilde{\mu}, m_1] + \\ & \quad - \frac{29}{12} g^2 g^2 Y_d^{i112} L_{F4,1,-2}[\tilde{\mu}, m_1] - \frac{1}{2} g^4 \tilde{\mu}^2 Y_d^{i112} L_{F4,2,-2}[\tilde{\mu}, m_1] - \\ & \quad - \frac{4}{3} g^1 g^2 Y_d^{i112} L_{F5,1,-3}[\tilde{\mu}, m_1] - \frac{1}{2} g^3 g^2 Y_d^{i112} L_{F3,1,-1}[\tilde{\mu}, m_2] + \frac{4}{3} g^2 Y_d^{i112} L_{F3,2,-2}[\tilde{\mu}, m_2] - \\ & \quad - \frac{1}{2} g^4 Y_d^{i112} (4 m_2^2 + 3 \tilde{\mu}^2) L_{F3,2,-1}[\tilde{\mu}, m_2] + \frac{4}{3} g^4 Y_d^{i112} L_{F4,1,-2}[\tilde{\mu}, m_2] - \\ & \quad - \frac{1}{2} g^4 Y_d^{i112} L_{F4,2,-3}[\tilde{\mu}, m_2] + \frac{3}{2} g^4 Y_d^{i112} (m_2^2 + \tilde{\mu}^2) L_{F4,2,-2}[\tilde{\mu}, m_2] - 4 g^2 Y_d^{i112} L_{F5,1,-3}[\tilde{\mu}, m_2] - \\ & \quad - \frac{1}{2} Y_{dPR} Y_d^{i112} Y_d^{i1r} Y_{uST} Y_u^{i1r} L_{F2,1,0}[\tilde{\mu}, \tilde{\mu}, m_u^r] + Y_{dPR} Y_d^{i112} Y_d^{i1r} Y_{uST} Y_u^{i1r} L_{F3,1,-1}[\tilde{\mu}, \tilde{\mu}, m_u^r] - \\ & \quad - \frac{1}{2} Y_{dPR} Y_d^{i112} Y_d^{i1r} Y_{uST} Y_u^{i1r} L_{F4,1,-1}[\tilde{\mu}, \tilde{\mu}, m_u^r] - Y_d^{i112} Y_{dPR} Y_{uST} Y_u^{i1r} L_{F2,1,0}[\tilde{\$$